Problem Set 3

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Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in R using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

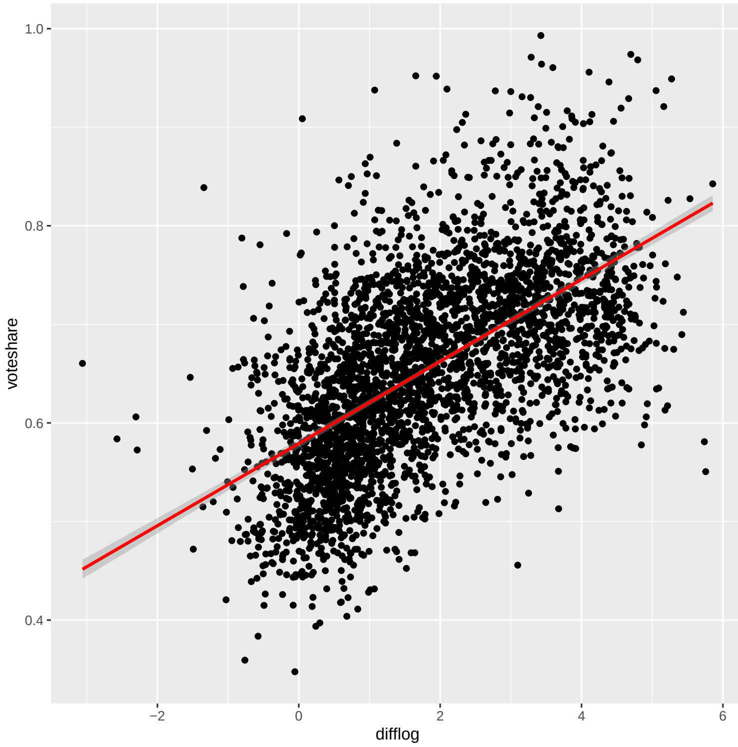
```r
regression1 <- lm(voteshare ~ difflog, data = inc.sub)
```

Table 1: Table of Coefficients

	Dependent variable:
	voteshare Regression 1
difflog	0.042***
	(0.001)
Constant	0.579***
	(0.002)
Observations	3,193
R^2	0.367
Adjusted R^2	0.367
Residual Std. Error	0.079 (df = 3191)
F Statistic	1,852.791*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

```r
scatter1 <-
  ggplot(inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point() +
  geom_smooth(method='lm',col="red")
```



3. Save the residuals of the model in a separate object.

```
residuals1 <- regression1$residuals
```

4. Write the prediction equation.

```
regression1
```

The prediction equation is y = 0.579 + 0.042x or voteshare = 0.579 + 0.042difflog in the context of this specific data.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

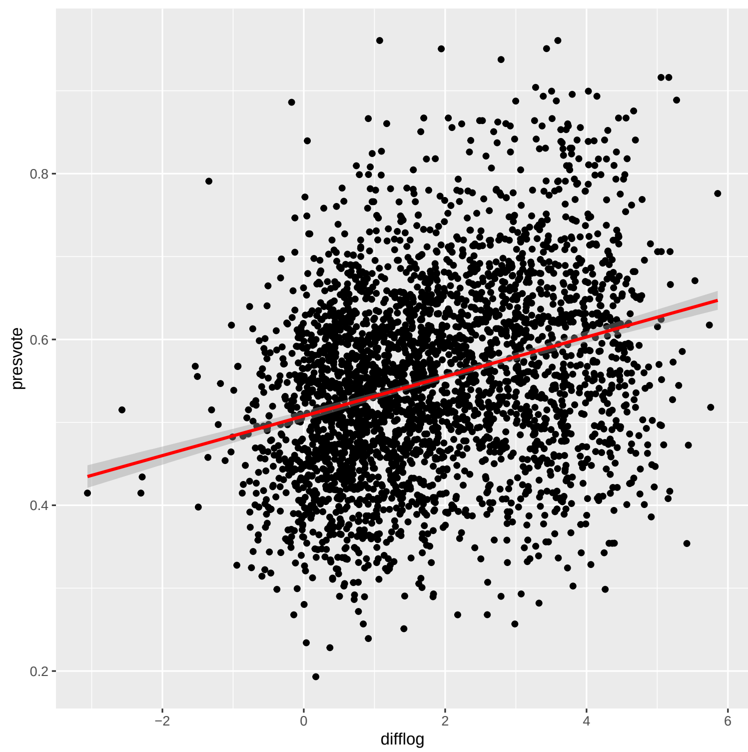
```
regression2 <- lm(presvote ~ difflog , data = inc.sub)
```

Table 2: Table of Coefficients

	Dependent variable:
	presvote Regression 2
difflog	0.024***
	(0.001)
Constant	0.508***
	(0.003)
Observations	3,193
R^2	0.088
Adjusted R^2	0.088
Residual Std. Error	0.110 (df = 3191)
F Statistic	307.715*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

```
scatter2 <-
  ggplot(inc.sub, aes(x = difflog , y = presvote)) +
  geom_point() +
  geom_smooth(method = 'lm', col = "red")
```



3. Save the residuals of the model in a separate object.

```
residuals2 <- regression2$residuals
```

4. Write the prediction equation.

```
regression2
```

The prediction equation is y = 0.508 + 0.024x or presvote = 0.508 + 0.024difflog in the context of this specific data.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

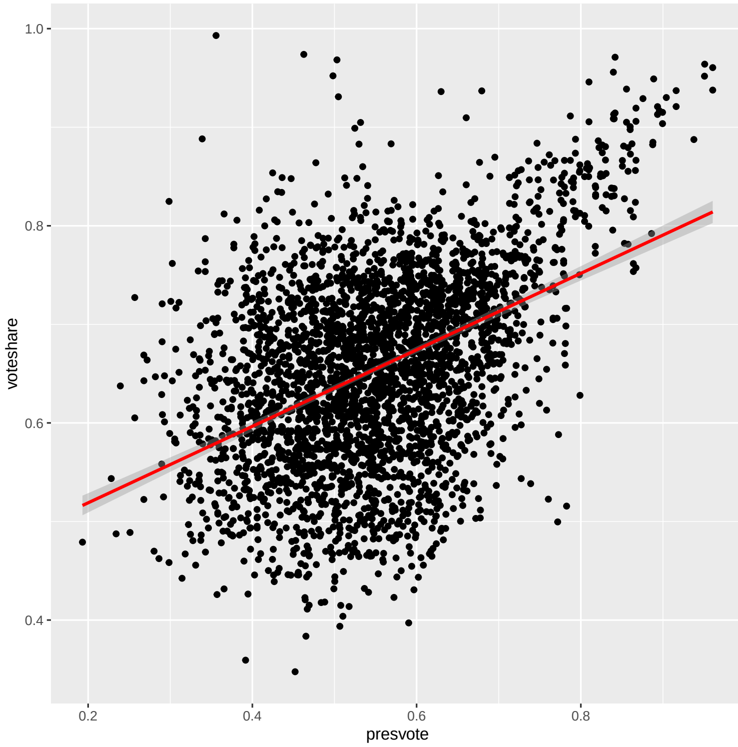
```r
regression3 <- lm(voteshare ~ presvote, data = inc.sub)
```

Table 3: Table of Coefficients

	Dependent variable:
	voteshare
	Regression 3
presvote	0.388^{***}
	(0.013)
Constant	0.441^{***}
	(0.008)
Observations	3,193
R^2	0.206
Adjusted R^2	0.206
Residual Std. Error	0.088 (df = 3191)
F Statistic	826.950^{***} (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

```r
scatter3 <-
  ggplot(inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point() +
  geom_smooth(method = 'lm', col = "red")
```



3. Write the prediction equation.

```
regression3
```

The prediction equation is y = 0.441 + 0.388x or voteshare = 0.441 + 0.388presvote in the context of this specific data.

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

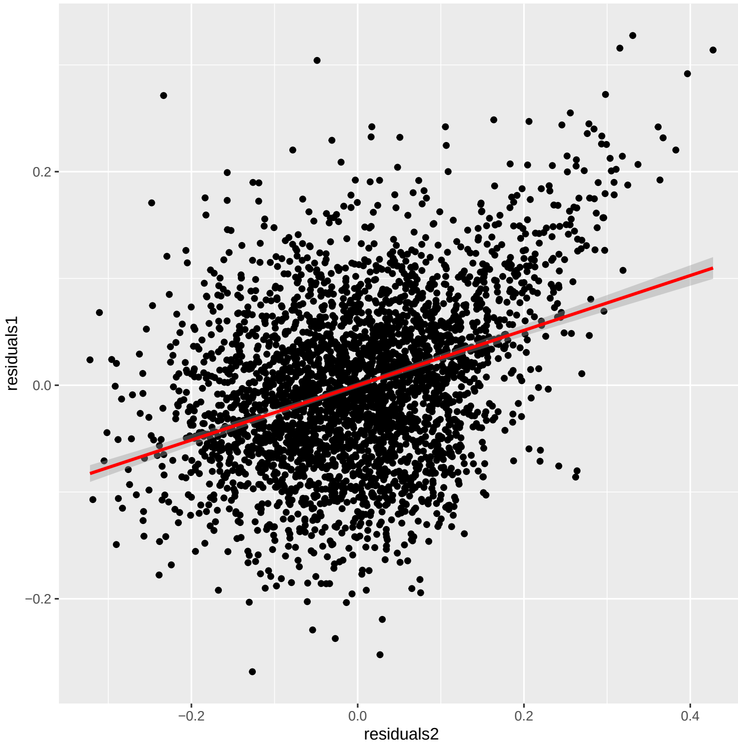
```r
resid_regression <- lm(residuals1 ~ residuals2)
```

Table 4: Table of Coefficients

	Dependent variable:
	residuals1 Residual Regression
residuals2	0.257***
	(0.012)
Constant	−0.000
	(0.001)
Observations	3,193
R^2	0.130
Adjusted R^2	0.130
Residual Std. Error	0.073 (df = 3191)
F Statistic	476.975*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two residuals and add the regression line.

```r
scatter4 <-
  ggplot(df1, aes(x = residuals2, y = residuals1)) +
  geom_point() +
  geom_smooth(method = 'lm', col = "red")
scatter4
```



3. Write the prediction equation.

```
resid_regression
```

The prediction equation is y = -1.942e-18 + 0.257x or residuals1 = -1.942e-18 + 0.257residuals2 in the context of this specific data. The constant for the equation was pulled from the summary of the regression due to its small size.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
regression4 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
```

Table 5: Table of Coefficients

	Dependent variable:
	voteshare Regression 4
difflog	0.036***
	(0.001)
presvote	0.257***
	(0.012)
Constant	0.449***
	(0.006)
Observations	3,193
R^2	0.450
Adjusted R^2	0.449
Residual Std. Error	0.073 (df = 3190)
F Statistic	1,302.947*** (df = 2; 3190)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Write the prediction equation.

```
regression4
```

The prediction equation is y = 0.449 + 0.036x_1 + 0.257x_2 or voteshare = 0.449 + 0.036difflog + 0.257presvote in the context of this specific regression.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

In this output and in the output in Question 4, the slope for presvote impact on voteshare is identical to the slope for residuals2 impact on residuals1, which are both 0.2569. These two values are the same as they both represent the effect of the vote share of the presidental candidate of the incumbent's party on the incumbent's vote share whilst controlling for the confounding effect of the difference in campaign spending between incumbent and challenger. This is due to the fact that the residuals from the regressions in question 1 and 2 (residuals1 and 2, respectively), represent the incumbent's vote share and vote share of the presidential candidate of the incumbent's party that is not explained by the difference in campaign spending between incumbent and challenger. When these two residuals are ran in a regression with one another, the slope of residuals2 effect on residuals1 represents the independent effect of presvote on voteshare unconfounded from difflog. Running a multiple regression of the effects of both presvote and difflog on voteshare, the coefficient for presvote will therefore be the same as residuals2 coefficient in question 4, as multiple regression coefficients statistically control for the effects of other variables.